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The Rise (and Fall?) of China’s Demand for Imported Timber: A Network Analysis of the International Roundwood Trade

Valentin Mathieu^{a,b,*}, David W. Shanafelt^b

^a *Université de Lorraine, AgroParisTech, INRAE, Silva, 54000, Nancy, France,*

^b *Université de Lorraine, Université de Strasbourg, AgroParisTech, CNRS, INRAE, BETA, 54000, Nancy, France,*

Abstract

The international trade in wood products is an increasingly complex, interconnected component of the global economy, shaped by environmental, geopolitical, competitive, and macroeconomic forces. While the literature offers comprehensive studies of the international timber trade, few examine the structure of the trade network itself, leaving its evolution poorly understood. To address this gap, we model global roundwood trade as a network of country-to-country flows from 1996 to 2023 and analyze two downstream products — sawnwood and wood-based panels — as separate networks to detect movements up the value chain. The roundwood network proved resilient: economic crises, the COVID-19 pandemic, trade disputes, and ecological disasters register only as short-term perturbations, never as structural breaks. Over the long term, however, it grew more interconnected and more polarized around a single demand hub: China, through which around 71% of total roundwood trade value flowed by 2021. This concentration is demand-side rather than supply-side: a fragmented export market, from which Russia has structurally withdrawn, set against demand centered on China. Dependence on a single buyer is the network’s chief vulnerability. China’s contribution has fallen since 2021, under geopolitical disruptions, domestic self-sufficiency policies, and structural economic shifts. The retreat reflects a broad contraction in its import demand, not a move up the value chain: its roundwood and sawnwood imports fell together. These findings point to an emerging post-China era, in which the most exposed exporters, such as New Zealand, face pressure to diversify — both by finding new markets and by moving up the value chain.

Keywords: Globalization, Trade, Wood products, Network theory, Resilience, Market concentration, Forest policy

*Corresponding author

Email addresses: valentin.mathieu@agroparistech.fr (Valentin Mathieu), david.shanafelt@inrae.fr (David W. Shanafelt)

1. Introduction

The international wood products trade stands as an increasingly complex and interconnected component of the global economy, tied to environmental concerns, geopolitical shifts, evolving competitive dynamics, and macroeconomic trends. Because forest resources are unevenly distributed and concentrated in a few countries (*e.g.*, Russia, Brazil, Canada, the United States, and China) (FAO, 2024b), countries engage in international trade to balance supply with domestic demand (Long et al., 2019; Huang et al., 2024).

International wood product trade accelerated substantially over the 21st century, under converging economic and technological pressures. The main drivers were sustained global economic growth, rising demand for construction materials fueled by urbanization, trade liberalization, and technological innovations that improved operational efficiency along the forestry value chain¹ (Prestemon et al., 2003; UNECE/FAO, 2022b). This acceleration has shifted deforestation pressures geographically: forest resources accumulate in nations with predominantly non-indigenous, plantation-based forest systems — typically developed economies with established silviculture infrastructure — while extractive pressure falls on regions where indigenous forest ecosystems face depletion, mostly in developing nations (Prestemon et al., 2003; Leblois et al., 2017; Pendrill et al., 2019; Abman and Lundberg, 2020). This spatial divergence highlights how trade-mediated demand drives deforestation and forest degradation.²

The acceleration of the international wood trade has also coincided with a growing environmental consciousness and a rising preference for “green” materials with carbon storage capacity. Driven by climate change mitigation goals, this shift has produced a range of environmental regulations and sustainability initiatives (Prestemon et al., 2003). The emerging forest-based bioeconomy, exemplified by strategies like the European Bioeconomy Strategy, aims to replace non-renewable resources with high-value-added products from woody biomass (Wolfslehner et al., 2016; Winkel and European Forest Institute, 2017). This creates a paradox: while increased wood demand can support decarbonization and sustainable development, unsustainable sourcing risks exacerbating deforestation and forest degradation.

Consequently, market-based tools like forest certification schemes³ promote sustainable practices, sometimes restricting wood products trade (Guan and Ip Ping Sheong, 2019; Chen et al., 2020; Boubacar and Sissoko, 2025). Furthermore, timber trade regulations, primarily enforced by developed countries, aim to exclude

¹These innovations encompass advancements in plantation management systems, forest harvesting operations, and manufacturing processes.

²Illegal logging and its associated trade also contribute to wider environmental damage — including global anthropogenic carbon dioxide emissions and loss of biodiversity — and deprive governments of billions of dollars in vital revenues, foster corruption, undermine the rule of law, and can fuel conflict (Lawson and MacFaul, 2010).

³Such as the Forest Stewardship Council (FSC) and the Program for the Endorsement of Forest Certification (PEFC).

illegal timber from domestic markets, limiting trade volumes (Moral-Pajares et al., 2020; Rougieux and Jonsson, 2021; Apeti and N'Doua, 2023; Kim et al., 2024).⁴ While tariff barriers have declined due to international trade agreements, such trade regulations illustrate that non-tariff barriers (NTBs) increasingly influence wood product trade (UNECE/FAO, 2022b).⁵ Notable trade disputes over wood products, such as those between China and the United States (Muhammad and Jones, 2021; Pan et al., 2021), Russian roundwood export restrictions (Turner et al., 2008; Solberg et al., 2010; Lin and Zhang, 2017; Guan and Yang, 2024), and the Canada–US softwood lumber dispute (Van Kooten and Johnston, 2014; Johnston and Van Kooten, 2017), underscore a tension between free trade principles and national policy objectives. Such measures, while often aiming for sustainability or industry protection, can disrupt trade, raise prices, and shift production and consumption patterns.

In recent years, economic and geopolitical disruptions, such as the COVID-19 pandemic and the war in Ukraine, have affected the global supply chain for wood products, leading to volatility in timber prices and shipping costs (FAO, 2024a; UNECE/FAO, 2022a). Climate change impacts on forest ecosystems may worsen such instability, increasing the frequency and severity of natural disturbances like storms, wildfires, and pest outbreaks (Seidl et al., 2011; Seidl and Rammer, 2017; Curtis et al., 2018; Tyukavina et al., 2022; Patacca et al., 2023), directly disrupting timber supply chains and market stability, and degrading wood quality (García-Jácome et al., 2025). This apparent vulnerability to external shocks suggests that trade resilience is not absolute but contingent on global economic, environmental, and political stability (García-Jácome et al., 2025; Ma et al., 2025).

Understanding the complex web of these relationships and trade flows is essential for policy formulation and strategic planning. The literature extensively studies the international timber market and trade, spanning scales from the global to the subnational and covering a range of wood products (*e.g.*, Müller et al., 2004; Raunikaar et al., 2010; Caurila et al., 2013; Van Kooten and Johnston, 2014; Buongiorno, 2015; Rougieux, 2017; Rougieux and Jonsson, 2021; Shen and Lovrić, 2022). These studies employ three primary perspectives: trade mechanism analysis, forecasting, and policy analysis (Buongiorno, 2016; Rivière and Caurila, 2020; Mathieu and Roda, 2023). Across these perspectives, traditional approaches draw primarily on two families of models rooted in economic theory — forest sector models and econometric models (including gravity models) (Buongiorno, 1996; Latta et al., 2013; Northway et al., 2013). Both families rely on nationally reported data and treat countries as the main unit of analysis (Amador and Cabral, 2017; Shen and Lovrić,

⁴Examples are the European Union’s Action Plan for Forest Law Enforcement, Governance and Trade (FLEGT), the European Union Timber Regulation (EUTR), the US Lacey Act, the Australia Illegal Logging Prohibition Act, or the Japan Clean Wood Act.

⁵NTBs include quotas, embargoes, economic sanctions, export bans, stringent environmental policies, and phytosanitary controls (Li et al., 2007; Sun et al., 2010; Buongiorno and Johnston, 2018; UNECE/FAO, 2022a; FAO, 2024a).

2022). However, neither family was designed to capture the structure of trade itself, and the two fall short in opposite ways. Forest sector models could represent bilateral trade in principle but rarely do in practice: in the most widely used global model, the Global Forest Products Model, each country exports to and imports from a single world market rather than named partners (Buongiorno, 2015), and regionalized variants resolve trade flows only among a few dozen aggregate regions (Kallio et al., 2004; Latta et al., 2013). Gravity models, by contrast, do explain bilateral flows, but one country pair at a time, from the two partners’ characteristics (Buongiorno, 2016); the structure the pairs form — which countries trade with which, directly or through intermediaries — is not itself an object of analysis (De Benedictis and Tajoli, 2011). Neither approach therefore represents the trade system as an interdependent whole (Schweitzer et al., 2009; Fagiolo et al., 2009), and it captures neither the cascading interdependencies of contemporary global value chains, nor the indirect trade linkages in which goods reach their final destination through third countries (De Benedictis and Tajoli, 2011; Amador and Cabral, 2017; Liu et al., 2025). Consequently, these approaches offer limited structural insight and may misjudge systemic properties such as supply chain fragility, contagion effects, and network resilience (Fèvre et al., 2006; Huang et al., 2024).

We address this gap by adopting a network-theoretic approach that considers the trade flows between countries as the unit of analysis (De Benedictis and Tajoli, 2011; Amador and Cabral, 2017). This perspective treats the complete set of bilateral flows as a single system, revealing the underlying organization and evolution of the wood product trade (Long et al., 2019; Shen and Lovrić, 2022). Indeed, network analysis has long been used to study complex global interactions, such as in the live animal trade to explain the spread of disease (Fèvre et al., 2006).

Network analysis has markedly advanced the understanding of global wood product trade dynamics, moving beyond traditional economic models to reveal structural characteristics, competitive dynamics, evolutionary trends, and both the resilience and the vulnerability of these networks to external shocks (Lovrić et al., 2018; Long et al., 2019; Gao et al., 2024; Huang et al., 2024). Previous research indicates that these trade networks have evolved and expanded in the 21st century, showing increased complexity and interdependence, broader participation, and enhanced overall trade efficiency and resilience (Wang et al., 2020; Gao et al., 2024; Huang et al., 2024; Liu et al., 2024). Network analysis has also identified key players and their evolving roles: historically, North American and European countries were central, but emerging economies like China have become dominant importers and increasingly central players across the supply chain (Zhou et al., 2021; Gao et al., 2024; Huang et al., 2024; Liu et al., 2024). Russia and Canada remain major timber exporters, and New Zealand is a significant raw material exporter (Zhou et al., 2021). Despite dynamic changes, these networks maintain a relatively stable core-periphery structure, with China and India now prominent in the core (Wang et al., 2020). While core countries boost network resilience, their dominance introduces vulnerabilities (Huang et al., 2024). Beyond structural descriptions, the literature has also investigated

factors influencing network evolution, including economic scale, geographical distance, cultural differences, and forest resource endowment (Gao et al., 2024), internal trade effects (reciprocity and transitivity) (Shen and Lovrić, 2022), and policy changes (Huang et al., 2024).

Current research on forest product trade networks is young and still largely exploratory: it has mapped the broad contours of the trade, but rarely with the depth needed to characterize any single product’s network in full. Most studies take a macroscopic view, often combining different wood product types, which can distort network understanding and lead to less effective trade policies (Barigozzi et al., 2010; Liu et al., 2024). Conversely, narrowing the lens to a single product invites the opposite blind spot: it conceals movements up the value chain, where a decline in the trade of unprocessed wood may mark a shift toward processed products — such as sawnwood and panels — rather than a genuine fall in wood trade overall (Richards et al., 2022). Capturing such movements requires analyzing raw and processed wood products together, disaggregated rather than pooled. Within this chain, the most consequential gap concerns its raw base: roundwood — unprocessed timber directly harvested from forests — whose trade network has yet to be comprehensively characterized, either structurally or over time. Four features make roundwood the natural focal product of such a joint analysis. First, roundwood is the chain’s common upstream input: the primary commodity of global forest product supply chains and the second most traded wood product by volume over the last decade.⁶ Second, roundwood supply is anchored to forest endowments — processing capacity can relocate, but forests cannot — so its trade geography reflects the uneven global distribution of forest resources more directly than that of any processed product. Supply-side shocks originate in this layer before propagating downstream. Third, roundwood stands among the wood products most prominently targeted by trade policy: log-export bans and taxes have been adopted by more than 30 countries to foster domestic processing (Forest Trends, 2022; Resosudarmo and Yusuf, 2006; Amoah et al., 2009; Marchand and Zerbo, 2025), Russia’s escalating restrictions being only the most consequential recent case (Turner et al., 2008; Solberg et al., 2010; Lin and Zhang, 2017; Guan and Yang, 2024).⁷ Policy interventions thus register first — and most sharply — in the roundwood trade network. Fourth, roundwood trade is the layer most immediately tied to sustainable forest management and deforestation; leaving its network under-characterized therefore limits the evidence available to forest policy.

The present work therefore characterizes the structure and temporal evolution of the global roundwood trade network over 1996–2023. To read roundwood’s trajectory against possible movements up the value chain, we additionally analyze its two main downstream products — sawnwood and wood-based panels — as separate networks rather than a pooled aggregate, keeping roundwood as the focal product. We characterize

⁶According to FAOSTAT data: <https://www.fao.org/faostat/en/#data/FO>.

⁷Even policies with no trade component surface there: China’s nationwide logging ban in natural forests was followed by a rise of some 15% in its solid-wood imports, with trade in other wood products largely unaffected (Zhang and Chen, 2021).

these networks through complementary quantitative metrics, drawing on graph theory alongside traditional trade analysis. We hypothesize that the global roundwood trade network is systematically concentrated in exports (reflecting concentrated forest endowments) while remaining structurally sensitive to exogenous disruption events, including economic crises, geopolitical instabilities, and natural disasters. This hypothesis comprises three propositions: (1) export concentration remains moderately high over the study period; (2) the network is systematically vulnerable to major disruption events; and (3) it recovers its structure in the short term following such events.

2. Material and methods

2.1. Bilateral trade data collection

We extracted bilateral trade data for 1996–2023⁸ from the UN Comtrade database,⁹ which compiles the annual trade reports of more than 200 countries, using the “comtradeapicall” 1.2.1 Python package ([untradestats, 2024](#)).

Traded commodities are classified under the Harmonized System (HS) nomenclature ([World Customs Organization, 1983](#)), a general scheme revised about every five years. The HS is ill-suited to a forest-sector analysis for two reasons: it is not sector-specific — wood products may be grouped with unrelated commodities such as bamboo or plastic furniture — and its revisions add, remove, or split commodity codes, creating discontinuities in the time series. We therefore mapped HS codes onto sector-specific forest-product categories using the 2022 FAO Classification of Forest Products ([FAO, 2022](#)), together with the HS revision documentation, ensuring consistency across editions. On this basis, we extracted and aggregated the flows of three products: roundwood (FAO division code 012),¹⁰ sawnwood (05), and wood-based panels (07). We analyze these as three separate networks rather than pooling them into a single “wood products” aggregate. Pooling would blur product-specific structure and, in particular, hide any shift from roundwood toward sawnwood and panels, which keeping the products distinct allows us to test.

Finally, we retained only import and export flows, discarded poorly specified flows (for example, “non-elsewhere specified” partners), and removed the 5% of flows with the lowest net weight or value as implausibly small.¹¹ The full extraction procedure and the HS-to-FAO correspondence across editions are detailed in supplementary material [A1](#).

⁸We ended the series in 2023 because Comtrade reports can be revised for up to two years after publication.

⁹Data were extracted in June 2026.

¹⁰The 2022 FAO Classification of Forest Products includes the following divisions under the section “Wood in the rough (roundwood)” (code 01): “Wood fuel (including wood for charcoal)” (code 011), and “Wood in the rough other than wood fuel” (code 012). Our work focuses solely on material usage (FAO division code 012) and excludes energy products.

¹¹In the case of roundwood, implausibly small trade flows correspond to those with a net weight below 145 kilograms and traded value below US\$348.

2.2. Two levels of aggregation: countries and the European Union

European Union (EU) member states form a deeply integrated single market, and a large share of their wood trade is exchanged among themselves — often between neighbors, where the nearest sawmill lies across a national border. A log from northern Sweden may cross into Finland, or one from Austria into Germany, rather than travel farther to a domestic mill. Such cross-border flows register as strong two-way bonds between member states, swelling intra-EU trade. Keeping each member as a separate trader therefore leaves a large volume of intra-EU flows in the dataset that can obscure the trade between Europe and the rest of the world. To address this, we process the bilateral trade flows at two levels of aggregation. At the country level, each country is kept as a distinct trader, as in the raw data. At the EU-aggregated level, we treat the EU as a single trader: we remove all intra-EU trade flows and, for each external trading partner, sum the individual member states’ flows into a single EU flow (separately for the exporter- and importer-reported values, so that mirror flows are preserved). In other words, a flow between the EU and a partner country is the sum of all member states’ individual flows with that partner.

We include 28 states in the EU aggregate.¹² This aggregate retains the United Kingdom despite its 2020 withdrawal and excludes Norway and Switzerland. Because EU membership expanded over the study period — most acceding states joined between 2004 and 2013 — applying a fixed 28-member aggregate across 1996–2023 is an approximation. We adopt it deliberately, to remove intra-EU flows consistently over time rather than to track the EU’s exact legal composition year by year.

We carry all three products and both aggregation levels through all subsequent steps of the analysis.

2.3. Trade network building from bilateral trade data

At each aggregation level, we represent the trade network for each product and year as an edge list — an alternative to the standard “adjacency matrix” representation. In network theory, an edge list is a mathematical representation of a network, which is made up of nodes, or vertices (*e.g.*, countries or individuals) and edges, or links (*e.g.*, the movement of goods or people between countries, or the exchange of information or money) (Albert and Barabási, 2002; Newman, 2003). An example of an edge list for a simple network is provided in Figure 1. In our case, the edge lists are $N \times 2$ matrices that correspond to a forest product trade network with N edges in year t . Each edge represents a trade flow (either export or import) between nodes (countries), from the country of origin (*i.e.*, the exporter, indexed o), to the country of destination (*i.e.*, the importer, indexed d), in year $t \in \{1996, 1997, \dots, 2023\}$. Each element of the edge list is a node representing a country that exported or imported the product in year t .

¹²The 28 states considered are: Austria, Belgium, Bulgaria, Croatia, Cyprus, Czechia, Denmark, Estonia, Finland, France, Germany, Greece, Hungary, Ireland, Italy, Latvia, Lithuania, Luxembourg, Malta, the Netherlands, Poland, Portugal, Romania, Slovakia, Slovenia, Spain, Sweden, and the United Kingdom.

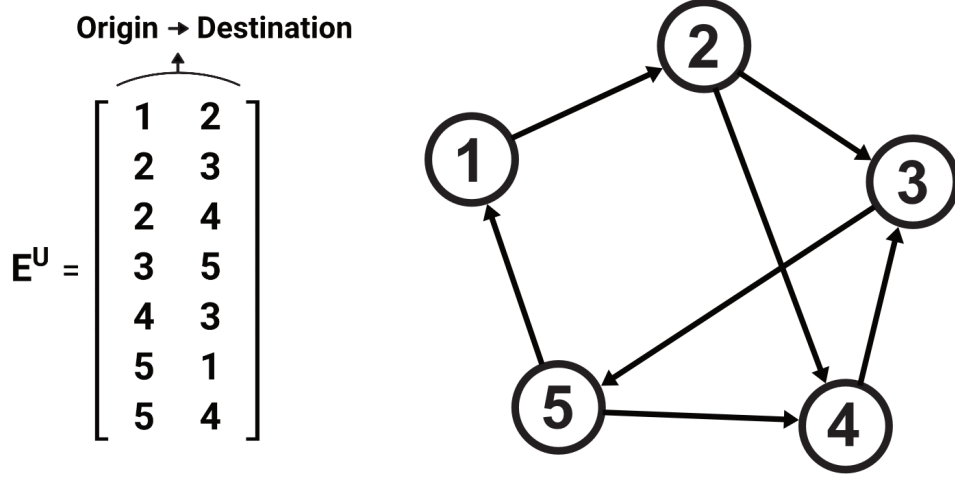


Figure 1: Illustration of an edge list, E^U , and its corresponding network graph for an *unweighted* and *directed* network of 5 nodes. In the edge list, each row corresponds to an edge from an origin (first column) to a destination (second column). In the network graph, circles indicate nodes and arrows the edges between them.

We treat each product-year network as weighted and directed: we record whether two countries trade, in which direction, and in what volume or value. In our case, we consider in parallel the traded value (in current US dollars) and the net weight (in kilograms) to be the “weight” of the trade flow.¹³ Due to data discrepancies, the reported trade value from the exporter and importer of the same trade flow may differ for at least two reasons: (1) importers report primary value as “cost, insurance and freight” (CIF), while exporters reports are expressed as “free on board” (FOB), which excludes CIF costs from trade value, leading to bilateral asymmetries; (2) one of the trading partners may not report the net weight or the value of trade (Gaulier and Zignago, 2010; Rougieux et al., 2017; Kallio and Solberg, 2018; Chen et al., 2022; Mitikj and Kaushik, 2024). To minimize the bias that such inconsistencies may cause, we consider both the exporter’s and importer’s reports of traded net weight and primary value: $X_{o,d} = (NetWeight_{o,d}^X, PrimaryValue_{o,d}^X)$ and $M_{o,d} = (NetWeight_{o,d}^M, PrimaryValue_{o,d}^M)$, respectively. We keep these two reports separate rather than reconciling them into a single value, as databases such as BACI do (Gaulier and Zignago, 2010); we detail this choice in supplementary material A4. Thus, for each product, the edge list corresponding to the weighted, directed trade network is a matrix of size $N \times 6$: each row represents a trade flow for a given year t , including the country of origin, the country of destination, and the trade values and net weights reported by the

¹³UN Comtrade also reports a “quantity” variable (qty) alongside net weight, which we set aside — not for lack of coverage (for roundwood it is recorded about as often as net weight) but because it is harder to use consistently: its unit is not common across our products (roundwood and sawnwood by volume, panels by mass), leaving no single volumetric unit for the analysis; it is more often flagged as a UN Statistics Division estimate than net weight; and, where both are reported, it implies a physically impossible density for about a quarter of roundwood flows. We therefore rely on net weight, reported consistently across all three products, alongside primary value (see supplementary material A2; Table A4).

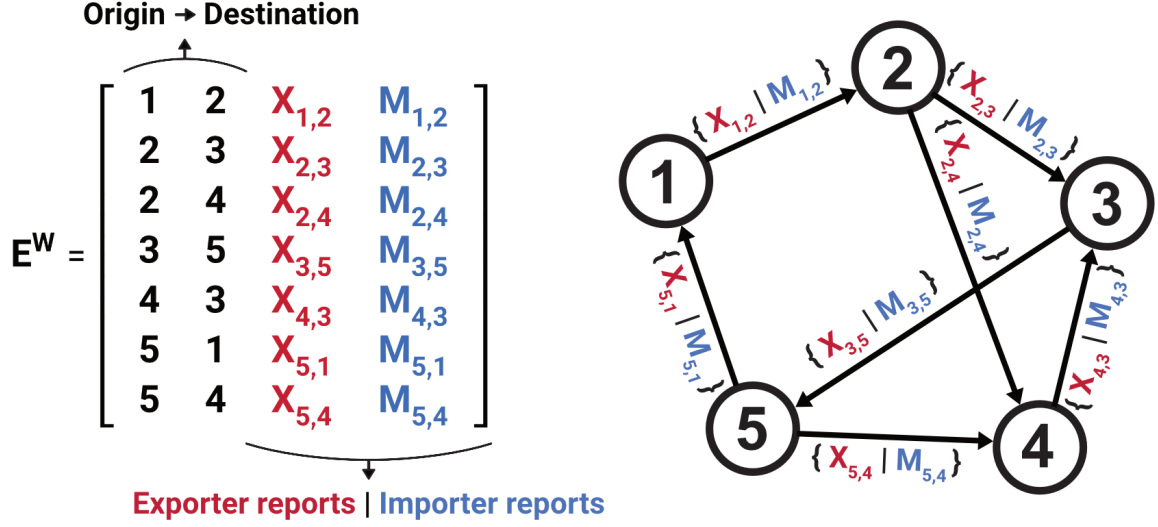


Figure 2: Illustration of an edge list, E^W , and its corresponding network graph for a *weighted* and *directed* trade network of 5 nodes. In the edge list, each row corresponds to an edge from an origin (first column) to a destination (second column) and include the report vectors of the country of origin (third column, red) and the country of destination (fourth column, blue). In the network graph, circles indicate nodes and arrows the edges between them.

exporter and importer, as shown in Figure 2.¹⁴ We define 28 edge lists to describe the trade network for one product, one for each year of trade.¹⁵

2.4. Trade network properties assessment

Edge lists and the raw network visualizations built from them are difficult to read on their own: neither the structure of a trade network nor its evolution over time can be inferred from them with any precision. Figure 3 makes the difficulty plain for roundwood — even reduced to its largest traders, the 2000 and 2020 networks are dense tangles of overlapping links, and comparing the two reveals little beyond the growth of China’s import bubble. To characterize this structure and track how it changes over time, we therefore computed a set of network metrics for each product and year, differentiating between exports and imports.¹⁶ We used the Python package “networkx” 3.4.2 (Hagberg et al., 2008) to compute these metrics.

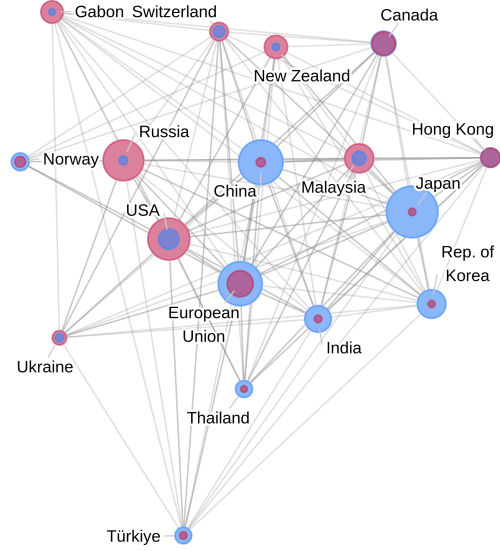
First, we assess the composition of the trade network. For each year, we compute the total number of nodes in the network, *i.e.*, the total number of trading countries. These countries are then assigned to one of three groups according to the share of their total reported traded value accounted for by exports, rather

¹⁴In fact, edge lists are close to mirror flows that can be derived from bilateral trade data, such as that provided by the UN Comtrade database.

¹⁵See Rayfield et al. (2011) and Thompson et al. (2017) for examples of this approach in ecological networks.

¹⁶Considering directed networks enables analysis from either the origin (exporter) or destination (importer) country’s perspective, allowing for the inference of exporter or importer trade behavior.

(a) Roundwood trade network in 2000



(b) Roundwood trade network in 2020

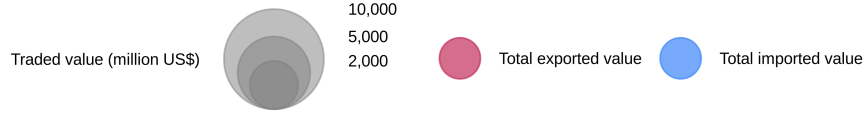
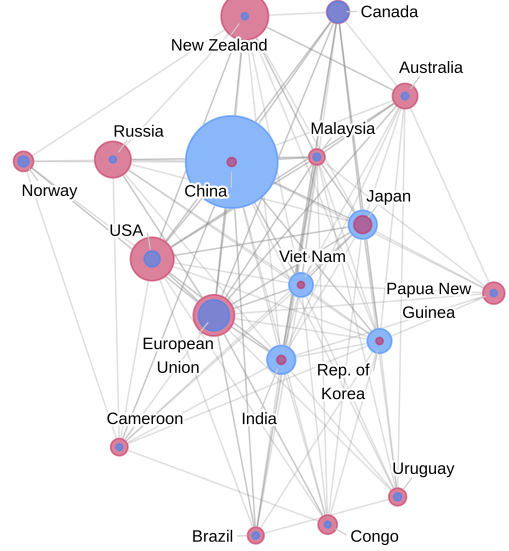


Figure 3: Roundwood trade networks in (a) 2000 and (b) 2020. Each node is a trading country; a gray line links two countries whenever either partner reported a trade flow between them. Node positions come from a force-directed layout (Kamada–Kawai) computed on the 2020 network and reused for 2000, so the two panels are comparable. Bubble area is proportional to traded value (million US\$), on a scale shared across both years; red bubbles show exported value, blue bubbles imported value. Only countries accounting for at least 1% of global export or import value in that year are shown. Network at the EU-aggregated level (intra-EU flows removed, EU treated as a single trader).

than by the mere direction of their flows. A country is a main exporter if exports represent at least 80% of its total reported traded value (reported exports plus imports), a main importer if imports represent at least 80%, and balanced otherwise. This value-based threshold separates countries by trade orientation: most countries both export and import at least once in a year, so a presence-based split leaves nearly all of them in a single undifferentiated group. These group counts, however, treat each year separately and cannot show how individual countries move within the network over time. To capture this, we build a trade profile for each country by cross-referencing its number of export and import partners (its out- and in-degree) with its reported export and import values — a positional reading that need not coincide exactly with the value-based classification above. A profile thus combines how widely a country trades — its trade integration — with how much it trades, in each direction. Because more than 200 countries trade each year, we profile

only the major contributors, *i.e.*, those accounting for at least 1% of the global export or import value — a selection for readability, not a filter on the analysis, since the metrics above are computed on the complete network. We then compare these profiles between an early and a recent year (2000 and 2020) to examine how the major traders’ roles have evolved.

Second, we characterize the network’s connectivity through the first four moments of the degree distribution — its mean, variance, skewness, and kurtosis (Albert and Barabási, 2002; Newman, 2003)— computed separately for exports and imports. The mean number of connections per node, or node degree, indicates how connected the network is overall. A high mean number of connections per node means that each country in the network has, on average, a high number of trading partners. The variance of the number of connections per node provides additional information about the dispersion of the number of connections. If the variance is close to zero, then the nodes in the network have more or less the same degree. A high variance indicates that the degree of connectivity of countries deviates widely from the mean. Skewness describes the balance between poorly and highly connected nodes. Negative skewness indicates a network dominated by highly connected nodes; positive skewness, one dominated by poorly connected nodes, with a thin tail of highly connected hubs. Zero skewness implies a symmetric distribution. Kurtosis provides information on how the degree distribution compares with a normal distribution in terms of its “tailedness” or “flatness”. A positive kurtosis¹⁷ indicates heavier tails than a normal distribution, meaning extreme outliers — countries with an unusually high or low number of trade relationships — occur more frequently than a Gaussian would predict. Conversely, negative kurtosis corresponds to lighter tails, where such extreme outliers are rarer and countries are spread more evenly across moderate distances from the mean.

Last, we assess market concentration with the traditional Herfindahl–Hirschman index (HHI), complementing the network metrics above. The HHI is defined as

$$HHI = \sum_{i=1}^N (MS_i)^2 \quad (1)$$

where N is the number of countries involved in trade and MS_i the market share of country i , *i.e.*, its share of total trade. We computed the HHI for exports and for imports, on both primary value and net weight. The HHI values can range from $1/N$ to 1. Based on these values, the US Federal Trade Commission distinguishes three market types (U.S. Department of Justice and the Federal Trade Commission, 2010):

- An HHI between 0.01 and 0.15 indicates an unconcentrated market, with no anti-competitive effects presumed;

¹⁷We consider Fisher’s definition of kurtosis. A Pearson’s definition would have necessitated the comparison of kurtosis values with 3 instead of 0.

- An *HHI* between 0.15 and 0.25 indicates a moderately concentrated market, with potentially substantial anti-competitive effects;
- An *HHI* above 0.25 indicates a highly concentrated market, with a presumption of anti-competitive effects.

To resolve market concentration to the level of individual countries, we computed each country’s annual contribution: the reduction in total trade that would result from removing the country, and all its flows, from the network. Equivalently, it is the share of total trade flowing through that country in a given year. The measure pinpoints the dominant traders, *i.e.*, those whose removal would cause the largest loss of trade. Because each flow is incident to two countries, these contributions sum to *200%*, not *100%*. Each flow carries two reported values — one from the exporter and one from the importer — that rarely coincide, owing to FOB-versus-CIF valuation and unilateral reporting (as described above). We therefore computed each country’s contribution from each report in turn, obtaining two estimates per country and year. We kept their mean as the country’s contribution and the gap between the two as a measure of reporting discrepancy: a wide gap flags a data-quality problem, which we preserve rather than reconcile away. This is the one metric for which we combine the two reports, and only for the plotted central line; the underlying pair is retained and shown as the band around it. We computed the contribution on both primary value and net weight. Trade value embeds unit prices as well as physical quantities, and wood prices vary across species and over time, so net weight offers a price-free check on the value-based results. We also examine these implied unit prices directly (see supplementary material [A6](#)). Both measures carry reporting limitations which we address, together with price effects, in the Discussion section.

Combining multiple complementary metrics offers a more comprehensive understanding of the network’s structure than any single metric alone can provide ([Shanafelt et al., 2017](#); [Salau et al., 2022](#)). For example, networks may have the same average number of connections but exhibit vastly different variances, resulting in markedly different topologies. Moreover, this set of metrics is sufficient for our purpose, since many network metrics are strongly correlated and add little independent information ([Baggio et al., 2011](#)). The metrics used and their interpretations are summarized in Table 1. Beyond these summary metrics, we also picture the bilateral partner structure they leave implicit — which trader ships roundwood to which — by rendering the network of a representative recent year (2020) as a chord diagram ([Figure 6](#)).

Table 1: List of the metrics used to describe the trade network and their interpretation.

Network metrics	Interpretation
Trade network composition	
Number of nodes	Number of countries involved in trade.

Network metrics	Interpretation
Number of main exporters	Countries whose exports account for at least 80% of their total reported trade value.
Number of main importers	Countries whose imports account for at least 80% of their total reported trade value.
Number of balanced countries	Countries for which neither exports nor imports reach the 80% threshold.
Country profiles	Number of trading partners cross-referenced with reported trade value, for major contributors (more than 1% of global exports or imports)
Trade network connectivity	
Mean number of connections per node	Average number of trading partners per country; an indicator of the network’s overall connectedness.
Variance in the number of connections per node	Variation in the number of trading partners relative to the mean.
Skewness in the number of connections per node	Proportion of low to highly connected countries in the network.
Kurtosis in the number of connections per node	Scarcity or abundance of countries that diverge from the mean connectivity.
Market concentration	
Herfindahl–Hirschman index	Concentration of the export and import markets, computed on both primary value and net weight.
Country contributions to trade	Share of total trade flowing through a given country, <i>i.e.</i> , the trade lost if it were removed; computed on both primary value and net weight.

2.5. Computational workflow and reproducibility

A sustainable data analysis workflow was undertaken using Snakemake 9.6.0 (Mölder et al., 2025). Snakemake is a workflow management system that ensures the reproducibility, adaptability, and transparency of the data analysis. This workflow is available on a GitHub repository,¹⁸ which provides access to the analysis and version control (Braga et al., 2023).

¹⁸GitHub repository address provided upon acceptance.

3. Results

Unless stated otherwise, the results presented below describe the roundwood trade network at the EU-aggregated level. The corresponding results for the country-level networks are reported in supplementary material A8, and for the sawnwood and wood-based panels networks at the EU-aggregated level in supplementary material A9 and supplementary material A10, respectively.

3.1. Composition of the trade network over time

The number of countries trading roundwood fluctuated little over the study period, averaging about 180 (Figure 4, panel a). It rose by 13% between 1996 and 2008 (from 166 to 187), held broadly stable until 2017, then fell by 8% to 172 countries in 2022, before rebounding by 6% to 182 in 2023. Short-term dips interrupted this trajectory — most visibly around 2008, 2016, 2019, and 2021 — and differed in size and direction across trader groups; we return to their possible drivers in the Discussion.

Throughout the period, main importers outnumbered main exporters, which in turn outnumbered balanced countries. Main exporters and main importers remained broadly stable in number (Figure 4, panel a), ranging from 53 to 71 and from 78 to 98 countries, respectively; balanced countries were fewer and varied more relative to their size (22 to 45). Main importers were somewhat more numerous at the end of the period than at the start (85 in 1996, 93 in 2023, up 9%), but this difference stayed within their year-to-year range and did not amount to a clear trend. The network’s composition was correspondingly stable (Figure 4, panel b): main importers made up 49% of trading countries on average (44–54% across years), main exporters 34% (28–39%), and balanced countries only 17% (13–25%). Specialized traders thus dominated the roundwood network: countries oriented mainly to exports or mainly to imports together accounted for about 83% of all traders, leaving balanced countries a small minority.

These counts describe how many countries traded in each role; they say nothing about how much trade each role carried. Most of the trade flowed through the two specialized groups (see supplementary material A5): main exporters, about 34% of trading countries, supplied on average 83% of the exporter-reported export value (69–88% across years), while main importers, about 49% of countries, absorbed 88% of the importer-reported import value (69–94%). Net weight gives the same picture (86% and 83% on average, respectively).¹⁹ That most trade flowed through the specialized groups does not by itself mean it was concentrated in a few hands: a group’s share can be spread across many countries. We take up that question next — first through the major traders’ profiles, then through the market-concentration indices.

¹⁹The two reporting sides of these totals did not coincide (*e.g.*, importer-reported value exceeded exporter-reported value in every year, and importer-reported net weight collapsed over the 2000–2006 window) — a discrepancy documented alongside the figures in supplementary material A5.

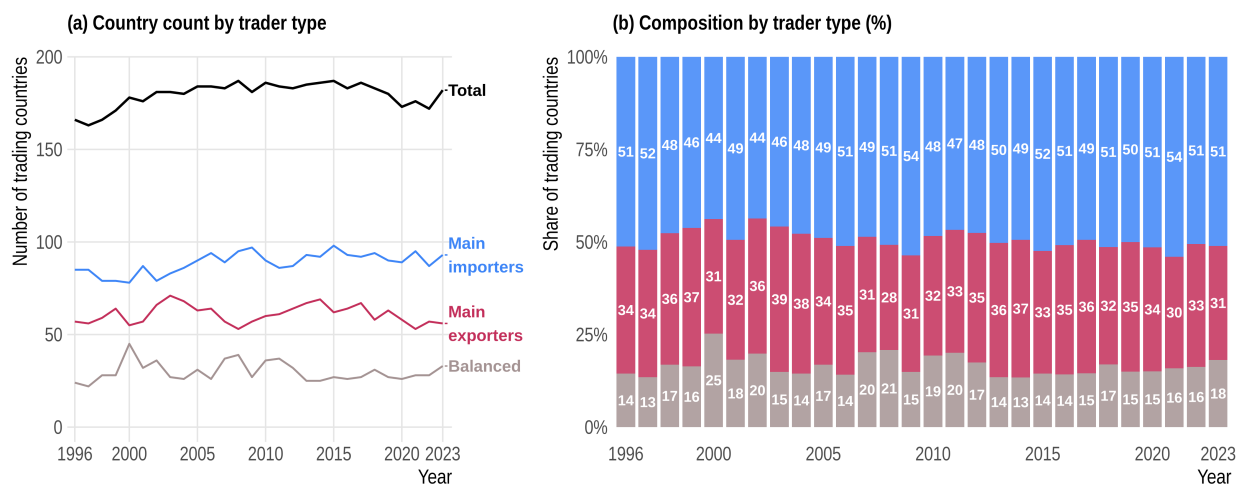


Figure 4: Composition of the roundwood trade network by trader type, 1996–2023: (a) the number of trading countries in each group and in total, and (b) each group’s share of all trading countries. Countries are classified by the share of their total reported trade value accounted for by exports — main exporters (red, exports $\geq 80\%$), main importers (blue, imports $\geq 80\%$), and balanced (gray) otherwise. Network at the EU-aggregated level (intra-EU flows removed, EU treated as a single trader).

Although 173 countries took part in the roundwood trade in 2020, only 18 — about 10% — each accounted for at least 1% of global export or import value (Figure 5), a first indication of the concentration that the following sections quantify. These countries fall into three roles, read at a glance from the diagonal of Figure 5: main exporters sit below it (more export than import partners), balanced traders on or near it, and main importers above it.²⁰

Main exporters sold to more partners than they bought from, but their reach varied widely: most had roughly 20–30 export partners, a few (such as Papua New Guinea and Uruguay) far fewer, while the United States stood apart with 74. Over the period the three largest suppliers were the USA, Russia, and New Zealand. Their ranks reshuffled between 2000 and 2020: New Zealand rose to become the world’s top supplier by value while holding a stable partner base, whereas the United States and Russia, though still major suppliers, lost reach. The United States shed 44% of its import partners and 9% of its export partners; Russia lost 7 export partners (21%). Former major exporters Ukraine and Gabon dropped out of the leading group, while Australia, Brazil, Uruguay, Papua New Guinea, Congo, and Cameroon entered it.

Balanced traders exported and imported in comparable value and connected to similar numbers of partners on each side. Such balance reflects the variety within the roundwood category rather than two-way trade in identical logs: a country can export one species or grade while importing another. The group was led by

²⁰This grouping follows each country’s position in the figure — its mix of export and import partners — and so need not coincide exactly with the value-based classification of the previous section.

the European Union — the most integrated trader in the network, with 157 partners in 2020 — alongside Canada, Norway, and Malaysia. Between 2000 and 2020 most balanced traders retreated in both value and partner count from their earlier positions, the European Union most visibly, and Switzerland left the leading group altogether.

Main importers, mostly Asian (China, India, Japan, the Republic of Korea, and Vietnam), bought from more partners than they sold to; their import connections were the most diversified in the network: China alone reached 76 partners, more than every supplier’s export reach except that of the United States. The importer hierarchy shifted decisively between 2000 and 2020: China overtook Japan as the dominant buyer of roundwood, its import value rising from \$1.7 billion to \$8.4 billion (+406%) while Japan’s fell from \$2.3 billion to \$562 million (−76%). As shares of global roundwood import value, China’s rose from about 20% in 2000 to 73% in 2020, while Japan’s fell from about 28% to 5%. Hong Kong dropped out, and Vietnam emerged with 46 import partners. China also widened its own buying, from 55 to 76 import partners, while India held steady and the Republic of Korea contracted slightly — the broadening of demand was concentrated in China and the new entrant rather than shared across Asia. China’s export partners more than doubled over the period (+139%, 25 new partners) even as its export value edged down 16% — a divergence we attribute to China’s sparse reporting of its own export flows; see the limitations in the Discussion and Conclusion section.

The trade profiles show how each major trader trades, and how much, but not with whom. Figure 6 completes the picture, drawing the 2020 flows among the ten largest traders as a chord diagram. It lays bare the network’s polarization: China’s arc dominates the circle, taking 73% of global roundwood import value, fed by four suppliers — New Zealand (23% of global export value), the United States (19%), the European Union (17%), and Russia (12%). New Zealand’s ribbon into China is the single thickest link, carrying 30% of China’s imports by value; only India and Japan (each about 4.9% of import value) register as other sizable buyers. Comparing the two panels previews a price effect we take up later: cheap, heavy logs from New Zealand and Russia occupy more of the circle by net weight (panel b) than by value (panel a) (see supplementary material A6).²¹ This China-centered structure is what the concentration indices and country contributions quantify in the sections that follow.

3.2. *Connectivity of the trade network over time*

The mean, variance, skewness, and kurtosis of the number of connections per country together describe how connections are distributed across the network (Figure 7). Three observations stand out for the mean. First, both exporters and importers grew more connected over the first half of the study period (1996–2008): the

²¹For legibility the main-text figure shows only the exporter-reported (free-on-board) side. The importer-reported (cost-insurance-freight) side is materially the same — China’s central position and its set of suppliers are unchanged, only the arc magnitudes shift with the reporting side — as the full four-panel version confirms (see supplementary material A7).

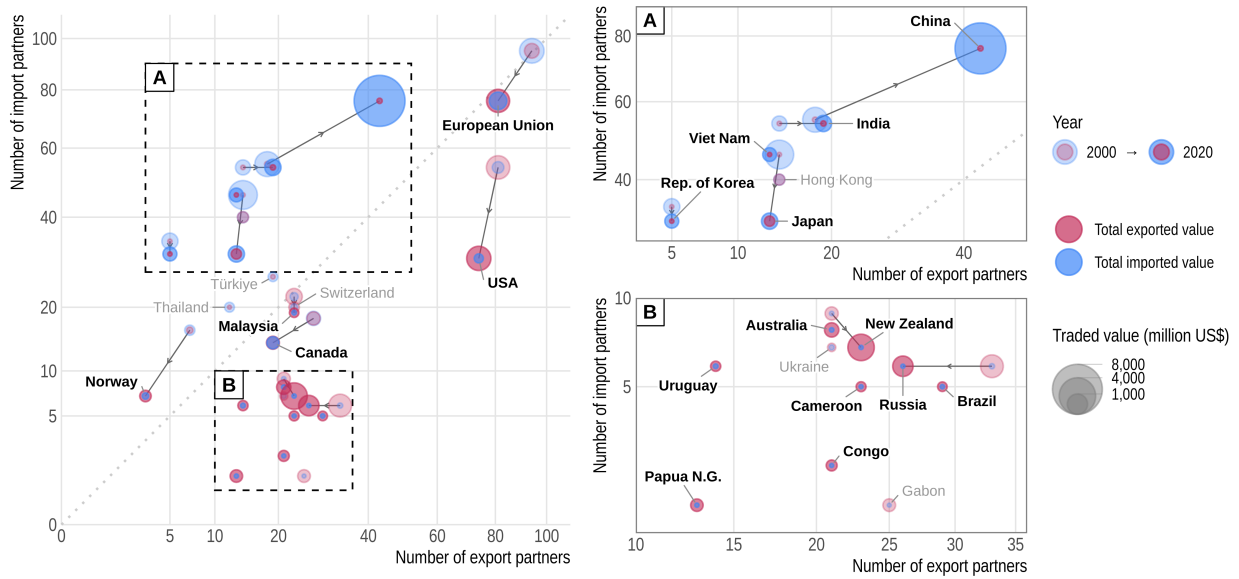
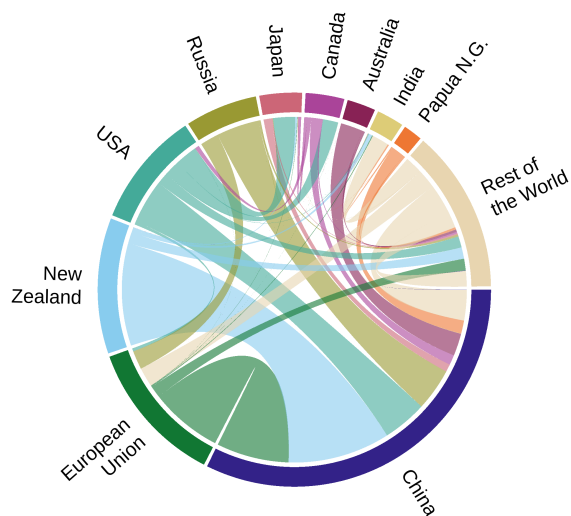


Figure 5: Trade profiles of the major roundwood traders, 2000 and 2020. Each country is positioned by its number of export partners (x-axis) and import partners (y-axis); bubble area is proportional to traded value (million US\$). Red bubbles show exported value, blue bubbles imported value; faded bubbles mark 2000 and solid bubbles 2020, joined by a trajectory. Insets enlarge two crowded regions: (A) the main Asian importers and (B) low-import exporters. Only countries accounting for at least 1% of global export or import value in 2000 or 2020 are shown. Network at the EU-aggregated level (intra-EU flows removed, EU treated as a single trader).

mean number of connections rose by 31% for exporters (from 7.5 to 9.9) and by 40% for importers (from 5.7 to 8.1). After 2008, mean connectivity declined for both, but more steeply for importers (21%, down to 6.3 in 2023) than for exporters (11%, down to 8.7). Second, exporters were on average 27% more connected than importers across the period (9 versus 7.1 connections), a gap that widened between the two halves (21% over 1996–2008, 31% over 2008–2023). Third, beyond these trends, mean connectivity fluctuated from year to year for both groups, with drops following 2008, 2016, 2019, and 2021.

The variance of the number of connections measures how far individual countries deviate from this mean. It is high for both exporters and importers. Each group comprises countries with a wide range of trading partners, some with far fewer connections than average (poorly connected countries) and others with far more (highly connected countries). Variance rose for both groups through the 2000s, peaking around 2008, before easing. Exporters and importers had comparable variance until the early 2000s, after which exporters' variance grew larger and the gap widened on average, with the same short-term fluctuations seen in the mean.

(a) Primary value — exporter reports (FOB, million USD)



(b) Net weight — exporter reports (tonnes)

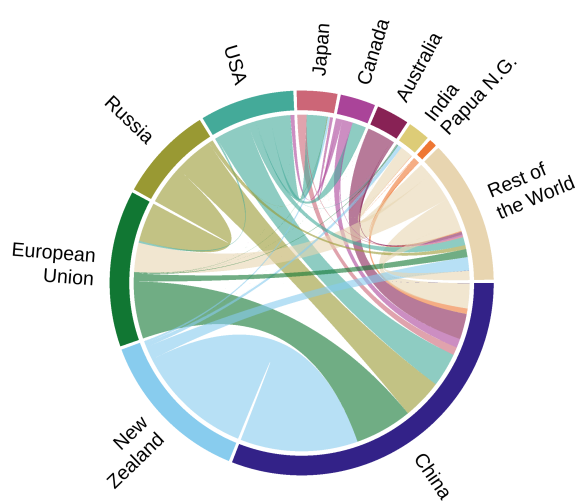


Figure 6: Chord diagram of the roundwood trade network in 2020, at the EU-aggregated level, from the exporters' reports: (a) primary value (free-on-board, million US\$) and (b) net weight (tonnes). Each arc is one of the ten largest traders by share of global export or import value; all other countries are pooled into “Rest of the World”. An arc's length is a trader's total trade (exports plus imports) on that measure, and a ribbon links an exporter to an importer with width proportional to the flow between them; ribbons take the exporter's color. The two panels use the same ten traders, ordered by involvement within each panel. The importer-reported (CIF) side is materially the same and is given, with the full four-panel version, in supplementary material [A7](#). Network at the EU-aggregated level (intra-EU flows removed, EU treated as a single trader).

Skewness in connections per country was positive in every year for both exporters and importers, indicating a preponderance of poorly connected countries alongside a thin tail of highly connected ones. As with the other metrics, it shows short-term, year-to-year fluctuations: the trend is stable for importers but fluctuates more for exporters, though exporter skewness changed little in absolute terms and stayed close to the importer values (both ranging roughly 3.5 to 4.7).

Kurtosis measures how heavy a distribution's tails are — how often values fall far from the mean relative to a normal distribution. It stayed positive throughout for both exporters and importers, and, read together with the positive skewness, this locates the excess weight in the upper tail: most countries fall below the mean connectivity, while a few highly connected hubs sit far above it. As with skewness, kurtosis shows short-term, year-to-year fluctuations: stable for importers and more variable for exporters, whose values also ran somewhat higher (both ranging roughly 14 to 27).

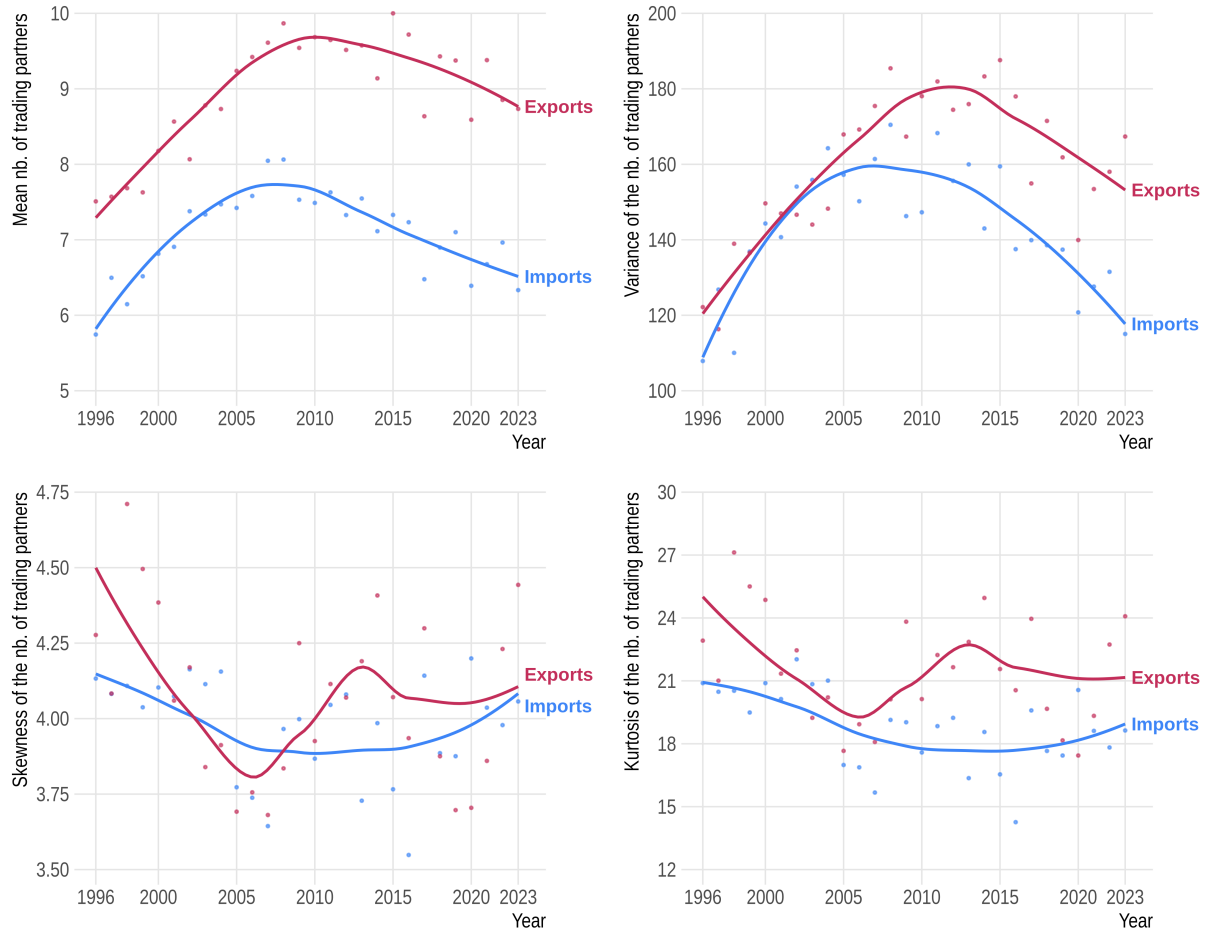


Figure 7: Distribution of the number of trading partners per country (node degree) in the roundwood trade network, 1996–2023: mean (top left), variance (top right), skewness (bottom left), and kurtosis (bottom right). Points are annual values; the line is a LOESS smooth highlighting the trend. Exports in red, imports in blue. Network at the EU-aggregated level (intra-EU flows removed, EU treated as a single trader).

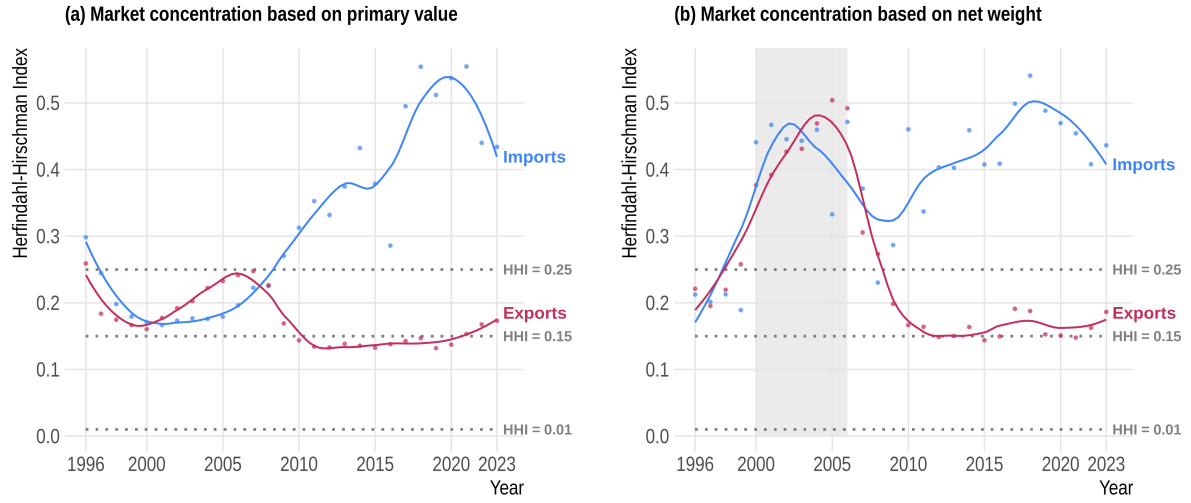


Figure 8: Herfindahl–Hirschman index (HHI) of the roundwood trade network, 1996–2023, computed on (a) primary value and (b) net weight. Exports in red, imports in blue; points are annual values, the line a LOESS smooth highlighting the trend. Dotted reference lines mark the thresholds between unconcentrated ($\text{HHI} < 0.15$), moderately concentrated ($0.15\text{--}0.25$), and highly concentrated (> 0.25) markets (US FTC). The gray vertical band in the net-weight panels marks the 2000–2006 window of unreliable net-weight reporting (see supplementary material A3). Network at the EU-aggregated level (intra-EU flows removed, EU treated as a single trader).

3.3. Market concentration in roundwood trade over time

Measured on primary value, roundwood trade started the period moderately concentrated and then diverged sharply between the two sides of the market (Figure 8, panel a). Concentration first fell on both sides between 1996 and 2000 — by 38% for exports (from 0.26 to 0.16) and 43% for imports (from 0.3 to 0.17) — easing toward an unconcentrated market. Exports then climbed back, approaching the high-concentration threshold of 0.25 by 2007 (0.25), before falling again and settling around the 0.15 boundary between unconcentrated and moderately concentrated markets (averaging 0.14 since 2010). Across the full period, however, exports remained moderately concentrated on average (0.17). Imports followed the opposite path. After a slow rise to 2006 (0.2), import concentration climbed steeply to a peak in the early 2020s (0.55) — well above the highly concentrated threshold — before easing back to 0.43 in 2023. Both series fluctuated year to year, most strikingly for imports between 2014 and 2016, when the index plunged from 0.43 to 0.29 before rebounding to 0.5 the following year. Over the study period, roundwood imports were thus mostly highly concentrated (averaging 0.42 since 2009), whereas exports stayed near the lower end of moderate concentration.

Computed on physical volumes rather than values, the net-weight HHI reflects concentration in the quantities traded and is unaffected by price changes (Figure 8, panel b). Early in the period, the two measures diverged: both exports and imports rose sharply to highly concentrated levels through the mid-2000s (peaking at 0.5

for exports in 2005 and 0.47 for imports in 2006). This surge warrants caution, however: it coincides with a cluster of missing (null) net-weight reports, likely tied to the Harmonized System revisions of 2002 and 2007 (see supplementary material [A3](#)). After 2007 both series fell steeply. Exports dropped to moderate concentration by 2010 (0.17) and held there (averaging 0.16 since 2010). Import concentration fell more briefly, stayed highly concentrated, and rose again after 2009 to a second peak in 2018 (0.54) before easing back (0.44 in 2023; averaging 0.43 since 2009). Setting aside the unreliable 2000–2006 window, the net-weight index tells the same story as the value-based one: exports moderately concentrated and stable, imports persistently highly concentrated.

Contributions from individual countries give a finer picture of market concentration (Figure 9). A country’s contribution is the share of the network’s total trade that would be lost if it and all its flows were removed — equivalently, the share that flows through it. We first describe the value-based contributions (left column of Figure 9) and turn to net weight below. Over the study period, the largest contributions came from eight traders (those above the 97th percentile of the annual cross-country distribution over 2019–2023; Figure 9). They fall into the same three groups as in the network composition (Figure 4): the main exporters (the United States, New Zealand, and Russia), two balanced traders (Canada and the European Union), and the main importers (China, Japan, and India).

Among the main exporters (Figure 9, panel a), the United States’ contribution declined slowly, from 40% in 1996 to 21% in 2023. New Zealand’s rose sharply but only recently: after holding near 6% through 2008, it climbed to 27% by 2023. Russia followed a more eventful path — rising from 16% in 1996 to a peak of 40% in 2007, falling back to 14% by 2013, holding around 15% until 2018, then collapsing to about 1% in 2023. This is consistent with the trade profiles (Figure 5), in which Russia lost both export partners and reach over the same period.

The two balanced traders followed contrasting paths (Figure 9, panel c): Canada’s contribution stayed broadly constant, averaging 9% across the period, whereas the European Union declined then recovered — from 18% in 1996 it declined through the late 2000s to 12% by 2012, before recovering steadily to 23% in 2023.

Among the main importers (Figure 9, panel e), Japan’s contribution fell almost without interruption — from 52% in 1996, the largest of any country at the outset, to a low of 2% in 2018 — before a modest rebound to 8% in 2023. India rose from 2% in 1996 to 20% in 2012, fell back to 5% by 2020, then edged up again to 9% in 2023. China’s rise dominated the period: from 5% in 1996 it climbed to 71% by 2021 — by then long the main importer, having overtaken Japan early in the period — before easing to 63% in 2023. Read together with Figures 5 and 8, this places around 71% of total trade value as passing through China in 2021 — on a measure in which each flow is counted at both of its endpoints, so that contributions sum to

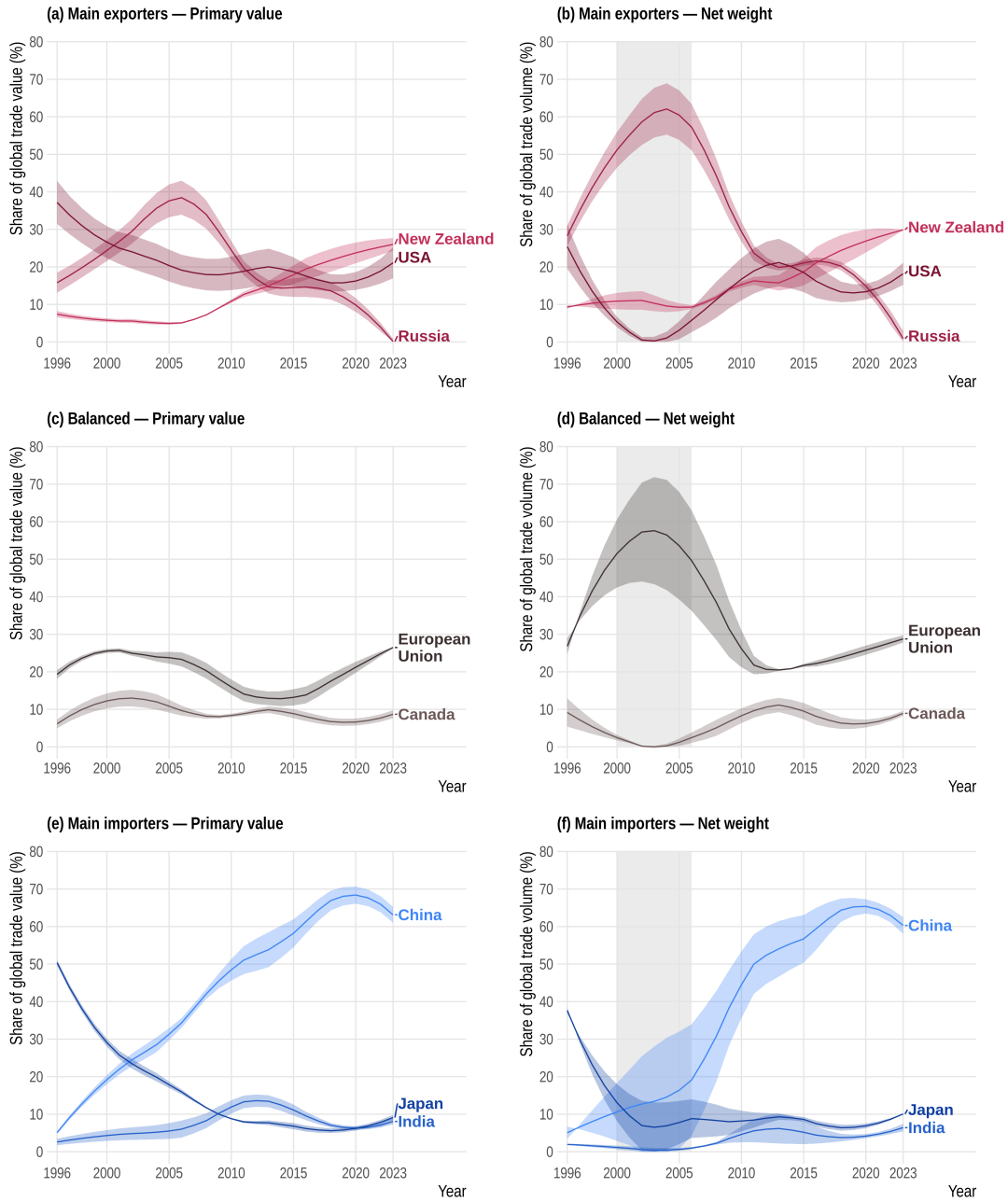


Figure 9: Each country's contribution to the roundwood trade network, 1996–2023, computed on primary value (left column) and net weight (right column). A country's contribution is the share of total trade lost if it and all its flows were removed; because each flow links two countries, contributions sum to 200%. Countries are grouped by trade orientation — main exporters (top row), balanced (middle), main importers (bottom) — and only those whose contribution exceeded the 97th percentile of the annual cross-country distribution over 2019–2023 are shown. The line is the mean of the exporter- and importer-reported contributions; the shaded band spans the two reports, so its width signals the discrepancy between them. The gray vertical band in the net-weight panels marks the 2000–2006 window of unreliable net-weight reporting (see supplementary material A3). Network at the EU-aggregated level.

200% — almost entirely on the import side, consistent with the high concentration of roundwood imports over the past decade.

Measured by net weight rather than value (right column of Figure 9), the same trends largely held: the United States and Japan declined, New Zealand and China rose, and Russia peaked in the mid-2000s before falling away. Two caveats apply. First, the 2000–2006 window cannot be trusted: over these years a cluster of missing net-weight reports (see supplementary material A3) mechanically inflated the remaining traders’ shares — Russia’s net-weight contribution averaged 59% over 2000–2006, against 32% by value. Second, several traders — chiefly the main exporters and the balanced European Union — contributed more by weight than by value; in 2023, for instance, New Zealand reached 34% by weight against 27% by value, and the European Union 25% against 23%. This gap is the expected signature of a price effect — cheap, heavy coniferous logs weigh more per dollar than the average flow — but it does not disturb the ranking of dominant traders, which value and net weight order almost identically (see supplementary material A6). Apart from Russia, whose contribution swung with short- to medium-term shocks, every major contributor showed a long-term structural movement over the study period.

4. Discussion and Conclusion

Contrary to our expectations, the global roundwood trade network proved substantially resilient to exogenous disruptions. Rather than registering major short-term shifts, it followed a long-term trend toward greater import concentration and polarization around China. Yet despite China’s marked growth in market power in recent decades, our findings suggest that its influence on the roundwood trade has declined in recent years, pointing to an impending shift toward a post-China hegemony era in the roundwood trade landscape.

The network structural metrics showed low to moderate sensitivity to shocks and disruptions. The number of main exporters, main importers, and balanced countries showed slight short-term drops over the studied period (Figure 4). Short-term increases and decreases in connectivity or market concentration metrics were also observed from year to year (Figures 7, 8). These rises and falls in the network structure likely result from three broad categories of events. First are economic disruptions, such as financial or banking crises (*e.g.*, the 2008 global financial crisis or the 2016 Chinese financial crisis) and supply chain disruptions (*e.g.*, the COVID-19 pandemic). Second are political events, such as trade disputes (Van Kooten and Johnston, 2014; Johnston and Van Kooten, 2017; Pan et al., 2021) or bilateral agreements, which affect trade relationships between countries. Third are ecological disasters. Hurricanes, tsunamis, and earthquakes can result in direct and indirect damage, as well as periods of reconstruction that can raise import demand or curtail export capacity. Forest fires and storms may affect the availability of roundwood, thereby altering exports and imports throughout the supply and production chains. These shocks left no lasting mark on the network

structure: each affected it for a year or so, after which the metrics returned to trend. This suggests that the trade network is resilient to exogenous shocks.

In fact, our results rather show that the roundwood trade network has followed long-term trends over the past decades. Its composition proved remarkably stable: the number of participating countries fluctuated little, and their division into main exporters, main importers, and balanced traders barely moved (Figure 4). What shifted over the long term was not this composition but the density of trade among these countries: connectivity tightened at the turn of the 21st century, as countries traded through more partners on both the export and import sides (Figure 7), consistent with broader globalization and trade intensification (Prestemon et al., 2003). Beneath this integration lies an uneven structure: a few highly connected countries (hubs) dominate and concentrate trade, while most of the network consists of poorly connected countries that barely contribute to roundwood trade volumes (Figures 5, 7). These hubs play a critical role in the network’s functionality (Huang et al., 2024) and can wield considerable market power, with potential anti-competitive effects on poorly connected countries, which constitute the majority of the trade network.

Throughout the study period, many major hubs in the international market, including the EU, the United States, Canada, and Russia, maintained a high number of trading partners. However, our findings highlight long-term shifts in the trade landscape over recent decades. We observed a decline among traditional core importers and exporters, including Japan, Russia, and the United States, concurrent with the rise of new key players like China and New Zealand (Figures 5, 9). Japan illustrates this decline. Between 1996 and 2023 its roundwood imports fell by about 87% in value — but so did its sawnwood (−69%) and wood-panel (−46%). The retreat spanned the whole value chain: Japan bought less wood of every kind rather than swapping processed imports for logs. This broad contraction is consistent with a prolonged fall in construction and an aging, shrinking population, both weakening domestic demand (Hashimoto et al., 2020; Cohen et al., 2000; Ministry of Agriculture, Forestry and Fisheries, 2011; Edgington, 2004), accompanied by rising self-sufficiency as post-war plantations matured (Ministry of Agriculture, Forestry and Fisheries, 2011; Aga, 2014). Only the pace differed by product: roundwood fell fastest, tilting the remaining imports toward processed wood — consistent with a decades-long move away from raw logs as Southeast Asian and Russian suppliers imposed export bans and taxes (Solberg et al., 2010; Tachibana, 2000; Moffett and Waggener, 1992). Especially, our findings show that China has become *the* central player in the roundwood trade, substantially increasing its number of import and export partners. By 2021, China’s imports accounted for 71% of the total roundwood trade value, leading to a highly concentrated and polarized import market (Figures 8, 9). This confirms findings from other studies (Ke et al., 2019; Long et al., 2019; Zhou et al., 2021; Shen and Lovrić, 2022; Huang et al., 2024).

This could appear as a paradox: although forest resources are geographically concentrated in a few countries,

our results show that the roundwood export market is not similarly concentrated (Figure 8). The relatively low export market concentration levels observed during the last decade are presumably due to Russia’s withdrawal from the export market initiated in 2007. The increase in export market concentration at the beginning of the 21st century coincided with Russia’s growing contribution to trade (Figures 8, 9). In 2007, Russian roundwood exports represented 45% of the global roundwood export value. Russia’s contribution to roundwood trade then fell from 40% in 2007 to less than 1% in 2023, over a period of successive export tax increases culminating in a complete export ban in 2022 (Figure 9) (Lin and Zhang, 2017; UNECE/FAO, 2022a,b; FAO, 2024b; Guan and Yang, 2024). These restrictions aimed less at halting exports than at capturing value domestically, moving Russia’s forest sector up the value chain toward domestically processed sawnwood and panels (Solberg et al., 2010) — a shift our data bear out (see supplementary material A12). Russia’s position as the main exporter of roundwood has since been overtaken by New Zealand, which accounted for 30% of total exports in 2023. In 2023, the next-largest roundwood exporters by value were the USA (23% of global export value), the European Union (15%), Canada (6%), Norway (4%), and Papua New Guinea (3%), followed by Japan, Brazil, Malaysia, and Uruguay, each near 2%. Such distribution of export value may explain the relatively low concentration levels of the roundwood export market.

In contrast, our findings suggest that the import market has become highly concentrated primarily due to demand from China (Figures 8, 9), which, paradoxically, also holds the world’s fifth-largest forest endowment and records the world’s largest annual net gain in forest area (FAO, 2024b). Indeed, sustained public spending together with market and policy reforms driven by forest-conservation goals have fueled a drastic expansion of China’s forest area — and especially of its planted forests — over recent decades (Démurger et al., 2009; Ke et al., 2019; Zhang, 2019; Zhao et al., 2022): planted forests have roughly doubled between 1990 and 2020 to around 90 million hectares — more than a third of the national forest area (Cheng et al., 2024). This expansion, however, has come largely through young, single-species plantations whose stand quality and growth remain low — a shortfall widely attributed to neglected silvicultural management, in particular limited tending and thinning (Hou et al., 2019; Ke et al., 2019; Hoffmann et al., 2018). This shortfall has held the mean annual increment of China’s plantations to about 4 m³ per hectare — some 56% below the world average for plantations under comparable growing conditions (Liu et al., 2019).²² China’s plantations

²²Reported growth rates for China’s plantations rarely exceed 10 m³ per hectare per year (Hou et al., 2019). The gap is clearest within a single genus: for eucalyptus — China’s fastest-growing plantation tree — mean annual increment runs only about 1.6–8.7 m³ per hectare per year, against 4–10 in West and Central Asia, 7–12 in South and Southeast Asia, 12–14 in North Africa and 15–70 in South America (Hoffmann et al., 2018); national inventory data echo this, China’s eucalyptus productivity rising only from 4.1 to 8.6 m³ per hectare per year between 1994 and 2018, with some 90% of stands still below 10 m³ per hectare per year (Zhang and Wang). The contrast with the world’s leading growers is starker still — Brazilian eucalyptus at 33–46 m³ per hectare per year (Stape et al., 2010), New Zealand radiata pine at 22–32 (Palmer et al., 2010) and intensively managed loblolly pine in the southern United States at 24–28 (350–400 ft³ ac-1 yr-1) (Fox et al.), all well above the roughly 4

are also lightly stocked, holding only some 35–53 m³ per hectare (Hou et al., 2019; Ke et al., 2019) — below the national forest average of about 89 m³ per hectare and far short of the world average of about 131 m³ per hectare (Ke et al., 2019). Nationally, China’s forests, both planted and natural, add only about 3.55 m³ per hectare a year, a quarter to a third of the rate in developed countries, and across 106 million hectares of young and middle-aged forest only 31% has reached half of its growth potential (Liu et al., 2019). With domestic production so constrained, timber consumption — primarily for industrial and construction uses — has outpaced domestic supply, making China highly dependent on imported wood products, which now account for approximately half of its total timber supply (Démurger et al., 2009; He and Xu, 2011; Hoffmann et al., 2018; Hou et al., 2019; Ke et al., 2019).²³

Such demand for imported roundwood fosters co-dependence between China and roundwood exporting countries. In fact, China serves as the main trading partner for most major exporters, making them highly reliant on its demand. For instance, in 2020, 46% of roundwood exports from the United States, 67% from Russia, 77% from Congo, 78% from Papua New Guinea, and 86% from New Zealand went to China. The case of New Zealand illustrates this co-dependence. Its rise as a major exporter is closely tied to China’s increasing demand and Russia’s concurrent withdrawal from the roundwood export market. The share of New Zealand’s roundwood exports to China surged from just 1.2% in 1996 to 90% in 2023. That exposure is the vulnerability itself: the more of its exports a country ties to one hub, the more a shift in that hub’s demand or policy can strand — so spreading sales across many destinations hedges against hub-specific shocks (Huang et al., 2024). The main roundwood suppliers fall along exactly this gradient (Figure 5). New Zealand is not alone at the dependent end: Papua New Guinea and Congo also send more than three-quarters of their roundwood to China through a handful of partners. The United States anchors the opposite pole, holding the widest export base of any single country — 72 to 93 partners — with under half its exports China-bound. The European Union reaches wider still, trading with 80 to 100 partners on each side; yet even its export value has lately swung to China — about 85% in 2020 — a measure of how hard the hub pulls, even on a broadly connected supplier. Diversification, then, is the lever that lowers this exposure, though not a free one: it trades the returns of riding the fastest-growing buyer for lower risk, and comes more readily to a large, diversified economy than a small, specialized one.

Overall, the concentration of imports around China’s demand, against a comparatively fragmented export

m³ per hectare per year of China’s plantations, which are dominated by slower-growing Chinese fir, larch and poplar — though such cross-species rates reflect differences in species, climate and rotation as much as in management; the strictly like-for-like comparison remains the World Bank benchmark in the main text, which holds growing conditions constant.

²³That dependence is not uniform across the value chain: China imports raw and semi-processed wood — logs and sawn timber — while exporting finished panels on a large scale, positioning it as the processing hub of the global wood-products chain (see supplementary material A11).

market, suggests that roundwood trade flows are predominantly demand-driven rather than supply-pushed. In the case of China, where wood is consumed primarily for industrial and construction purposes (Hou et al., 2019), this echoes the role of population and urbanization in shaping demand for materials (Steinberger et al., 2010), which has been identified in the literature (UNECE/FAO, 2022b; Mathieu and Roda, 2023; Villamor et al., 2024). Urbanization, through the growth, maintenance, and operation of buildings, infrastructure, and durable goods, is a major determinant of material consumption (Fernández, 2007; Hu et al., 2010; Krausmann et al., 2017; Deetman et al., 2020), which may have a considerable impact on roundwood trade. Regions undergoing population increase and urbanization — such as South and South-East Asia and, above all, sub-Saharan Africa, where most of the world’s future urban growth is projected to occur (UNECE/FAO, 2022b; Deetman et al., 2020) — are becoming centers of gravity that attract roundwood trade flows, with potential shifts in the role of countries within the trade network.

Most interestingly, China’s contribution to the roundwood trade began to decline after 2021 (Figure 9), potentially signalling a new phase in the global roundwood market. Several circumstances bear on this decline. Three plausibly drive it — external geopolitical disruptions, domestic self-sufficiency policies, and domestic structural economic changes — while a fourth, a movement up the value chain, finds little support in our data. The first and most immediate is external trade shocks. Most notably, Russia’s 2022 roundwood export ban sharply cut China’s imports of Russian logs: Russia, China’s fourth-largest supplier in 2021 at 7% of its roundwood imports, supplied just 3% by 2023 (see supplementary material A12). This shock was likely exacerbated by the still-ongoing 2018 US–China trade war (Muhammad and Jones, 2021; Pan et al., 2021). The United States’ share of China’s imports edged up from 9% in 2021 to 11% in 2023 — an arithmetic consequence of steeper declines among other suppliers, Russia’s collapse above all, rather than genuine substitution toward American wood: in absolute terms, China’s imports from the United States themselves fell by 29% over the same years. Given the current geopolitical climate, a Russian ban lift is unlikely, and escalating US–China trade barriers are expected to cause further marked trade disruptions and diversions in the medium term (Attinasi et al., 2024; World Bank, 2025; World Trade Organization, 2025).

Yet this retreat was too broad to be a supply story alone. Between 2021 and 2023, China’s roundwood imports nearly halved in value (down 45%), falling not from Russia alone (down 76%) but from every major supplier — the European Union (59%), New Zealand (34%), and the United States (29%). The drop is as large by net weight (38%) as by value, so it reflects a real contraction in volume, not a mere price effect (see supplementary material A11). A decline this general points less to external disruption than to forces within China itself, which we consider next.

The second is more policy-driven: China’s strategic ambition to enhance self-sufficiency, meeting more of its demand from domestic supply chains rather than imports (FAO, 2021; UNECE/FAO, 2022b; Attinasi et al.,

2024). Heavy dependence on imports, particularly as sourcing grows more concentrated, presents geopolitical risks for China. In this regard, China announced its “Plan for the Construction of the National Reserve Forest Project” in 2018. This project aims to foster a net increase of 200 million m³ per year by 2035 from 20 million newly planted hectares of national reserve forest — an amount designed to cover China’s current total net imports of forest products. This domestic focus aims to make the country less reliant on imports and mitigate sourcing risks.

The third is domestic: China’s declining contribution in the roundwood trade network may also reflect economic crises and structural change. Since the turn of the decade, China has faced a real estate crisis which, coupled with slowed productivity growth, an aging population and high debt levels, has substantially slowed its domestic demand for roundwood (World Bank, 2025). The contraction in construction — a primary outlet for wood — is striking: new residential floor space started fell from a peak of about 1.67 billion m² in 2019 to about 0.69 billion m² in 2023, a decline of nearly 60% (National Bureau of Statistics of China, 2020, 2024).²⁴ This slump again echoes the role of urbanization in fueling demand for materials such as wood. Furthermore, while China’s position in global value chains has been fueled by labor-intensive industry development (Su et al., 2020), increases in labor costs and investments in research and development toward advanced technologies have led China to pivot from consumer goods production and exports to technology-intensive capital production and exports over the last decades (FAO, 2021; Attinasi et al., 2024). In fact, labor-intensive consumer goods production is currently moving out of China and into countries such as Vietnam (FAO, 2021; Attinasi et al., 2024).

The fourth — a movement up the value chain, with China importing (semi)-processed wood products, such as sawnwood or wood-based panels, in place of raw logs — can be advanced, but in our data it does not account for the recent decline. Log-export bans give it a concrete channel, and Russia is the textbook case. After its January 2022 roundwood ban, China became a leading buyer of Russian plywood²⁵ and, more

²⁴Numbers refer to the National Bureau of Statistics of China indicator *Floor Space of Houses Newly Started, Residential Buildings* (in *Investment in Real Estate Development, January–December*), originally reported in units of 10,000 m². The corresponding all-buildings series — residential, office, and commercial premises combined — fell from 2.27 billion m² in 2019 to 0.95 billion m² in 2023, a decline of about 58% that closely matches the residential figure and confirms the contraction was broad-based.

²⁵Russian hardwood-plywood exports nearly halved over 2022–2023 (down 47%), to about 6% of the US\$11.8 billion global market, as the EU ceased imports and the United States slowed sharply; China and other non-sanctioning buyers absorbed the remainder at falling prices (Scott, 2024). Our data confirm this from China’s side: its reported imports of Russian plywood nearly tripled, from about US\$47 million in 2021 to US\$134 million in 2023, even as Russia’s total plywood exports more than halved — so China’s share of those exports climbed from roughly 2.5% to 16% by value, and to 21% by net weight. The weight share runs higher because China bought at a discount (about US\$0.82 versus a US\$1.04/kg world average in 2023), matching the falling prices Scott (2024) reports. Even so, the flow stays under 1% of the roughly US\$15 billion in annual global plywood trade.

broadly, of Russian wood-based panels as Western markets closed to it: China-reported imports of Russian wood-based panels roughly doubled, from US\$80 to US\$160 million between 2021 and 2023, even as Russian logs were withheld. This bilateral shift does not generalize. Had China climbed the value chain, its imports of semi-processed wood products (sawnwood and panels) should have risen as its log imports fell. They did not — or not to any degree that matters. Over 2021–2023, China’s roundwood imports fell by about 45% in value and 40% in net weight; its sawnwood imports fell too, by about 13% and 9%; and its panel imports — small throughout, since China is a net exporter of panels rather than a buyer — edged up only marginally (about 1% and 3%). Network contributions tell the same story: China’s sawnwood and panel shares have drifted down from mid-2010s peaks rather than up (see supplementary material [A9](#), Section [A10](#), and [A11](#)). China did move up the value chain, but earlier and by a different route than the one posited here. Through the 2000s and early 2010s it became the world’s dominant wood processor — drawing in growing volumes of both logs and sawnwood and exporting finished panels on a vast scale — rather than replacing its log imports with processed-wood imports. That build-out had largely run its course by the mid-2010s. The one climb still under way is Russia’s own, which we read as a redirection of processed wood under its export ban and its exclusion from Western markets, with China absorbing that output rather than itself driving the shift (see supplementary material [A12](#)). China’s recent withdrawal from the roundwood trade therefore reads less as a continued move up the value chain than as a broad contraction in its demand for imported wood.

In light of these factors, China’s declining integration into the roundwood trade network is likely to continue.²⁶ A potential “post-China” era in the global roundwood market raises a crucial concern for major exporters such as New Zealand: the need to find new, sustainable markets and so diversify away from Chinese demand ([Villamor et al., 2024](#)). But geography is only one axis of adjustment; the other is vertical — moving downstream to capture added value at home, exporting processed wood (sawnwood, panels, and beyond) rather than raw logs, so the same timber leaves in higher-value form. New Zealand has run the other way: its export boom was built on unprocessed logs, whose share of its wood-export value climbed from about 37% in 2000 to 73% in 2023 — the reverse of Russia’s climb up the value chain. Processing more at home would loosen its dependence on raw-log demand as surely as new markets would. But that dependence has deepened, not eased, so we raise added value as a strategic avenue, not a forecast.

By drawing on several complementary metrics rather than a single index, we obtained a fuller diagnosis of the network’s structure and its evolution ([Shanafelt et al., 2017](#); [Salau et al., 2022](#)). In particular, measuring each country’s contribution to traded value complements traditional concentration indices such as the Herfindahl–Hirschman index by resolving aggregate concentration to the level of individual traders

²⁶And, with its sawnwood imports and panel exports also easing from their mid-2010s peaks, the retreat is not confined to logs but spans the wood-products chain (see supplementary material [A9](#), and [A10](#)).

and their relative market power. More fundamentally, taking the trade flows themselves as the unit of analysis brought out structural features that the dominant tools of forest-trade analysis are not designed to capture. Forest sector models represent each country’s trade with an aggregate world market rather than the bilateral links among countries (Buongiorno, 2015), and gravity models explain those bilateral flows from the trading partners’ attributes one pair at a time (Buongiorno, 2016); neither represents the trade system as an interdependent whole (Schweitzer et al., 2009; Fagiolo et al., 2009). Our network view surfaced three such features: the strongly uneven, hub-and-spoke topology characteristic of trade webs (Fagiolo et al., 2009), here polarized around China’s import demand; each country’s systemic weight, measured as the trade that would be lost if it were removed from the network (Albert et al., 2000); and the network’s resilience, *i.e.*, its capacity to absorb shocks with only short-term perturbation before recovering, consistent with the resilience of core-heavy trade networks (Huang et al., 2024). Network analysis therefore does not replace gravity or forest sector models but supplies the system-level layer they leave out.

Network analysis is only as reliable as the trade data beneath it (Lovrić et al., 2018; Wang et al., 2020; Zhou et al., 2021; Huang et al., 2024). The two measures we use to weight trade flows — traded value and net weight — are each imperfect, and they fail in different ways. Value in current US dollars is sensitive to price effects, as roundwood prices vary markedly by species and over time. At the flow level this variation is large, but it does not map onto the trader groups we compare — their median prices are nearly identical — and each country’s share of trade value closely tracks its share of net weight (see supplementary material A6), so the two measures rank traders alike and price effects do not distort the group-level results. The exporter’s and importer’s figures also sit on different valuation bases — free-on-board and cost-insurance-freight, respectively — so the two are not directly comparable; and, being nominal, both embed inflation across years. Net weight sidesteps prices but depends on moisture content — pronounced for roundwood — and is distorted by reporting breaks around the Harmonized System revisions: it was largely left unreported (null) for roundwood and sawnwood over 2000–2006, and over an earlier window in the late 1990s for wood-based panels (see supplementary material A3). A further physical measure exists in the data — the reported quantity, largely in cubic metres — and would suit roundwood in particular. But it has problems of its own: it lacks a common unit across the three products, is often estimated, and, for a sizeable minority of flows, is physically inconsistent with the recorded net weight (see supplementary material A2). Both measures are further affected by incomplete reporting, where a flow is recorded by only one partner — more often for net weight than for value. Because value and net weight are vulnerable to different distortions, we report both throughout: where they agree, a pattern is unlikely to be a measurement artifact, and where they diverge, the divergence itself flags one. Similarly, we do not reconcile the exporter’s and importer’s reports into a single value, as databases such as BACI do (Gaulier and Zignago, 2010): the gap between them is an informative signal, not noise to remove (see supplementary material A4). Carrying both sides

lets the network absorb isolated gaps — a flow missed by one partner is often captured by the other — while exposing systematic discrepancies such as inconsistent reporting and outliers (Kallio and Solberg, 2018; Chen et al., 2022). China’s roundwood exports illustrate this. Between 2000 and 2020 China exported to 25 more countries, yet its own reported export value fell by about 16% (roughly US\$1.2 million; Figure 5) — a combination that is hard to credit as a real trend. China discloses almost none of its own roundwood export flows: about 72% went unreported in 2000 and nearly all (98%) by 2020, and the share of export value covered by its own reports fell from about 70% to 33%. Resting on so small and shrinking a share of the trade, this figure is too incomplete to read as a trend. Such cases counsel caution when interpreting bilateral trade data and point to harmonized, reconciled databases as complementary tools when the objective is the most accurate flow estimates rather than, as here, a structural and data-quality diagnosis (Gaulier and Zignago, 2010; Rougieux et al., 2017; Mathieu, 2023; Guo et al., 2026).

Beyond how trade flows are measured and reconciled, the unit of analysis is itself a modeling choice, and we treat it as one. Many European countries show a balanced trade profile, exporting roughly as much roundwood as they import. Such balance need not mean a country ships out and takes back the same logs: within a category as broad as roundwood, a trader can export one species or grade — domestic hardwood logs, say — while importing another, such as softwood or lower-grade logs for construction and pallets. For European traders the pattern is compounded by geography, as neighboring members meet demand across shared borders, leaving most of their trade exchanged among themselves. We therefore ran the analysis at two levels: with each country as a separate trader (see supplementary material A8), and with the European Union treated as a single node, its internal flows removed. Aggregating the EU strips out these intra-European flows and gives a clearer view of Europe’s trade with the rest of the world, better reflecting its collective weight alongside actors such as China, the United States, and Russia. More generally, aggregating countries into regions trades spatial resolution for a clearer reading of the network’s external structure, and the appropriate level depends on the question — a tension we make explicit by presenting both levels and flagging where they diverge.

Network analysis captures the trade system’s complexity and describes its organization, but it does not explain what produces that organization. Understanding the determinants and dynamics of trade structure remains open, as does consolidating the diverse set of available network metrics into a unified indicator system. Our parallel analysis of the roundwood, sawnwood, and wood-based-panel networks already reads them through a value-chain lens — testing, for instance, whether China’s retreat from logs was taken up by processed-wood trade, and finding it was not (see supplementary material A9, Section A10, and A11). Multi-layer networks would formalize this: by explicitly linking each product layer according to value-chain logics (Baggio et al., 2016; Barigozzi et al., 2010), they could turn our product-by-product reading into a single, integrated supply-chain analysis.

We close by returning to the three propositions we began with. The first — that export concentration would remain moderately high — held only in the letter. Export concentration did stay moderate (a Herfindahl–Hirschman index averaging 0.17), but it ran consistently below what the geographic concentration of forest endowments would predict and drifted toward the unconcentrated boundary over the last decade as Russia left the log-export market (Figure 8); the concentration that shaped the network was demand-side, centered on China, not supply-side.

The second proposition — that the network would show systematic vulnerability to major disruptions — did not hold. Neither the global financial crisis, the COVID-19 pandemic, trade disputes, nor ecological disasters lastingly reshaped the network; each showed up as a short-term perturbation, not a structural break (Figures 4, 7, 8). We found resilience where we expected fragility — though not invulnerability: the network’s polarization around a single hub is a concentrated, latent weakness, a dependence on that hub rather than the diffuse fragility the proposition envisaged.

The third — short-term recovery following disruptions — held, and it is the flip side of that resilience: the metrics that dipped after a shock returned to trend within a year or so, so what the network shows is recovery, not lasting damage. Our structural hypothesis was thus two-thirds borne out, failing only where it expected fragility. The reframing is the paper’s central finding: the roundwood network is resilient to external shocks but polarized around a single source of demand, so its future turns less on the next shock than on China — whose post-2021 retreat suggests that shift may already be under way.

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